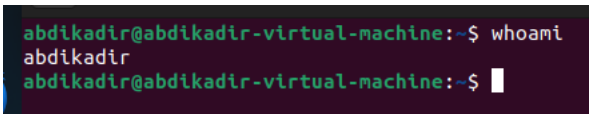
Linux Commands Notes

Chapters One to Four

Chapter One: Basic Commands

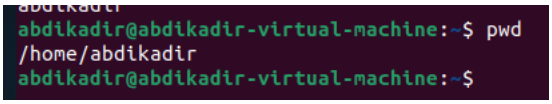
1. `whoami`

Definition: Shows the username of the user who is currently logged in.



2. `pwd`

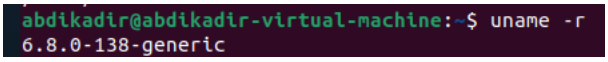
Definition: Print Working Directory. Shows the full path of the folder you are in now.



3. `uname -r`

Definition: Shows the version of the Linux kernel that is running.

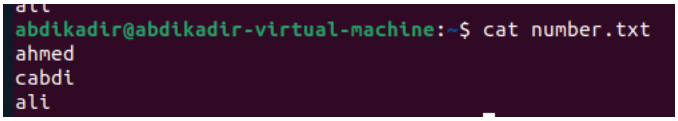
Test in terminal:



4. `cat`

Definition: Displays the file that describes the operating system (name and version).

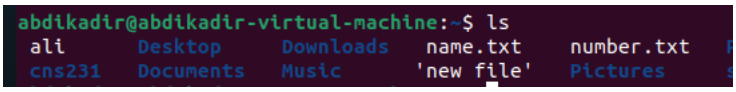
Test in terminal:



5. `ls`

Definition: Lists the files and folders in the current directory.

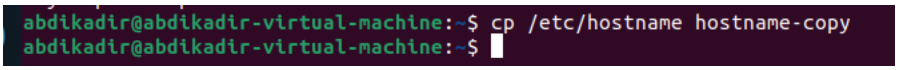
Test in terminal:



6. `cp`

Definition: Copies a file or folder. Format: cp source destination.

Test in terminal:



7. `mv`

Definition: Moves a file or folder, or renames it. Format: mv source destination.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mv hostname.copy hostname.moved
```

8. `apt`

Definition: Package manager for Ubuntu/Debian. Used to install, update and remove software.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ apt list --installed | head -5
```

9. `yum install…`

Definition: Package manager command used on Red Hat, CentOS and Fedora. It is NOT on Ubuntu, so on Ubuntu you will see "command not found".

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ yum install nano
```

10. `Ctrl + Alt + T`

Definition: Keyboard shortcut that opens a new Terminal window in Ubuntu. Press the three keys together.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ (press Ctrl + Alt + T)
```

Chapter Two: Files, Folders and Help

Navigation

11. `pwd`

Definition: Shows the full path of your current directory.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ pwd
```

12. `cd`

Definition: Change Directory. With no folder name it takes you back to your home directory.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd
```

13. `cd /etc`

Definition: Goes to the /etc folder, where system configuration files are kept.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd /etc && pwd
```

14. `cd ..`

Definition: Goes up one level, to the parent directory.

Before testing: Run cd /etc first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd ..
```

15. `cd ~`

Definition: Goes to your home directory (the ~ symbol means home).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd ~
```

16. `cd -`

Definition: Goes back to the previous directory you were in.

Before testing: Run cd /etc, then cd ~ first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd -
```

17. `cd /home/student/Downloads`

Definition: Goes to a folder using its full (absolute) path. Replace "student" with your user: use /home/abdikadir/Downloads.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd /home/abdikadir/Downloads
```

18. `cd ../Downloads`

Definition: Goes to the Downloads folder using a relative path from a sibling folder.

Before testing: Run cd ~/Documents first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd ../Downloads
```

19. `cd ../..`

Definition: Goes up two levels at once.

Before testing: Run cd ~/Documents first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd ../..
```

20. `cd ~/Documents`

Definition: Goes to the Documents folder inside your home directory.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd ~/Documents
```

21. `cd .`

Definition: Stays in the same directory (. means "here").

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd . && pwd
```

22. `cd /var/log`

Definition: Goes to the folder where system log files are stored.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd /var/log && pwd
```

23. `cd /`

Definition: Goes to the root directory, the top of the whole file system.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cd / && pwd
```

Listing files (ls)

24. `ls`

Definition: Lists files and folders in the current directory.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls
```

25. `ls /`

Definition: Lists everything inside the root directory.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls /
```

26. `ls /home`

Definition: Lists the home folders of all users.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls /home
```

27. `ls /sys/class/net`

Definition: Lists the network interfaces of the computer (for example lo, ens33).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls /sys/class/net
```

28. `ls -a`

Definition: Lists all files, including hidden ones (names that start with a dot).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -a
```

29. `ls -l`

Definition: Long format: shows permissions, owner, size and date.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -l
```

30. `ls -lh`

Definition: Long format with human-readable sizes (KB, MB).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -lh
```

31. `ls -lah`

Definition: Long format, human-readable sizes, and hidden files together.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -lah
```

32. `ls -l /etc`

Definition: Long listing of the /etc folder.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -l /etc
```

33. `ls -a ~`

Definition: Lists all files, including hidden ones, in your home directory.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -a ~
```

34. `ls /usr/bin/ls`

Definition: Shows that the ls program itself is a file stored in /usr/bin.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls /usr/bin/ls
```

35. `ls /usr/bin/mkdir`

Definition: Shows that the mkdir program is a file stored in /usr/bin.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls /usr/bin/mkdir
```

36. `ls -l /home`

Definition: Long listing of the /home folder.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -l /home
```

37. `ls -la`

Definition: Same as ls -l -a: long format plus hidden files.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -la
```

38. `ls --all`

Definition: Long option name for -a. Shows hidden files.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls --all
```

39. `ls --human-readable`

Definition: Long option name for -h. Used with -l to show readable sizes.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -l --human-readable
```

40. `ls --all --format=long`

Definition: Long option names combined: all files shown in long format.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ pwd
/home/abdikadir
abdikadir@abdikadir-virtual-machine:~$ cd Desktop/
abdikadir@abdikadir-virtual-machine:~/Desktop$ cd
abdikadir@abdikadir-virtual-machine:~$ cd ..
abdikadir@abdikadir-virtual-machine:/home$ cd/
bash: cd/: No such file or directory
abdikadir@abdikadir-virtual-machine:/home$ cd /
abdikadir@abdikadir-virtual-machine:/$ ls
bin   cdrom  etc   lib    lib64   lost+found  mnt  proc  run   snap  swapfile  tmp  var
boot  dev    home  lib32  libx32  media       opt  root  sbin  srv   sys       usr
abdikadir@abdikadir-virtual-machine:/$ ls -a
.   bin   cdrom  etc   lib    lib64   lost+found  mnt  proc  run   snap  swapfile  tmp
..  boot  dev    home  lib32  libx32  media       opt  root  sbin  srv   sys       usr
abdikadir@abdikadir-virtual-machine:/$ ls -l
total 3991632
lrwxrwxrwx   1 root root          7 Sag 22 21:07 bin -> usr/bin
drwxr-xr-x   4 root root       4096 Sag 22 21:37 boot
drwxrwxr-x   2 root root       4096 Sag 22 21:10 cdrom
drwxr-xr-x  19 root root       4080 KIT 10 10:34 dev
drwxr-xr-x 129 root root      12288 Tob  7 15:39 etc
drwxr-xr-x   3 root root       4096 Sag 22 21:12 home
lrwxrwxrwx   1 root root          7 Sag 22 21:07 lib -> usr/lib
lrwxrwxrwx   1 root root          9 Sag 22 21:07 lib32 -> usr/lib32
lrwxrwxrwx   1 root root          9 Sag 22 21:07 lib64 -> usr/lib64
lrwxrwxrwx   1 root root         10 Sag 22 21:07 libx32 -> usr/libx32
drwx------   2 root root      16384 Sag 22 21:06 lost+found
drwxr-xr-x   3 root root       4096 Sag 23 14:44 media
drwxr-xr-x   2 root root       4096 Lab 20  2024 mnt
drwxr-xr-x   2 root root       4096 Lab 20  2024 opt
dr-xr-xr-x 351 root root          0 KIT 10 10:31 proc
drwx------   4 root root       4096 Sag 22 21:39 root
drwxr-xr-x  33 root root        900 KIT 10 11:29 run
lrwxrwxrwx   1 root root          8 Sag 22 21:07 sbin -> usr/sbin
drwxr-xr-x  11 root root       4096 Lab 20  2024 snap
drwxr-xr-x   2 root root       4096 Lab 20  2024 srv
-rw-------   1 root root 4087349248 Sag 22 21:07 swapfile
dr-xr-xr-x  13 root root          0 KIT 10 10:31 sys
drwxrwxrwt  17 root root       4096 KIT 10 11:16 tmp
drwxr-xr-x  14 root root       4096 Lab 20  2024 usr
```

Variables and history

41. `echo $SHELL`

Definition: Prints which shell you are using (usually /bin/bash).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ echo $SHELL
```

42. `echo $PATH`

Definition: Prints the list of folders Linux searches when you type a command.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ echo $PATH
```

43. `echo $HOME`

Definition: Prints the path of your home directory.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ echo $HOME
```

44. `echo $USER`

Definition: Prints your username.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ echo $USER
```

45. `history`

Definition: Shows the list of commands you typed before.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ history
```

46. `echo $NAME`

Definition: Prints the value of the variable NAME. It is empty if NAME is not set yet.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ echo $NAME
```

```
abdikadir@abdikadir-virtual-machine:/$ echo

abdikadir@abdikadir-virtual-machine:/$ echo $Desktop

abdikadir@abdikadir-virtual-machine:/$ echo number.txt
number.txt
abdikadir@abdikadir-virtual-machine:/$ history
    1  ls
    2  ls-l
    3  ls -l /home/
    4  q
    5  ls /etc/
    6  ls lss
    7  l
    8  ls /temb/
    9  man
   10  ls /man/
   11  /man
   12  man man
   13  man ls
   14  ls
   15  history
   16  pwd
   17  cd/etc
   18  cd/document
   19  cd/documents
   20  q
   21  clear
   22  cd Desktop/
   23  cd
   24  pwd
   25  cd Documents/
   26  pwd
   27  Documents$ ls
   28  Documents$ ls-L
   29  clear
   30  mousepad
   31  sudo apt
   32  mousepad
   33  nano test1.txt
```

47. `export NAME=value`

Definition: Creates a variable called NAME and makes it available to programs you start.

Test in terminal:

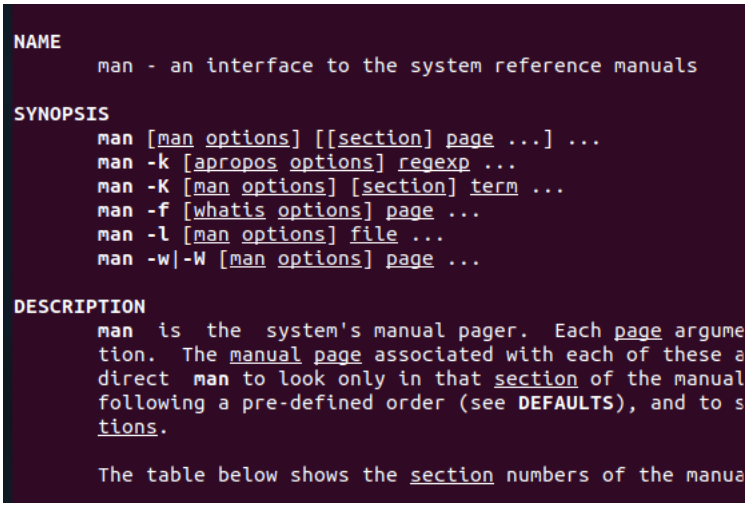
```
abdikadir@abdikadir-virtual-machine:~$ export NAME=abdikadir && echo $NAME
```

Getting help

48. `man ls`

Definition: Opens the manual page of ls. Press q to quit.

Test in terminal:



49. `man 1 passwd`

Definition: Opens section 1 of the manual for passwd (the command). Press q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ man 1 passwd
```

50. `man 5 passwd`

Definition: Opens section 5 of the manual for passwd (the file format of /etc/passwd). Press q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ man 5 passwd
```

51. `man rmdir`

Definition: Opens the manual page of rmdir. Press q to quit.

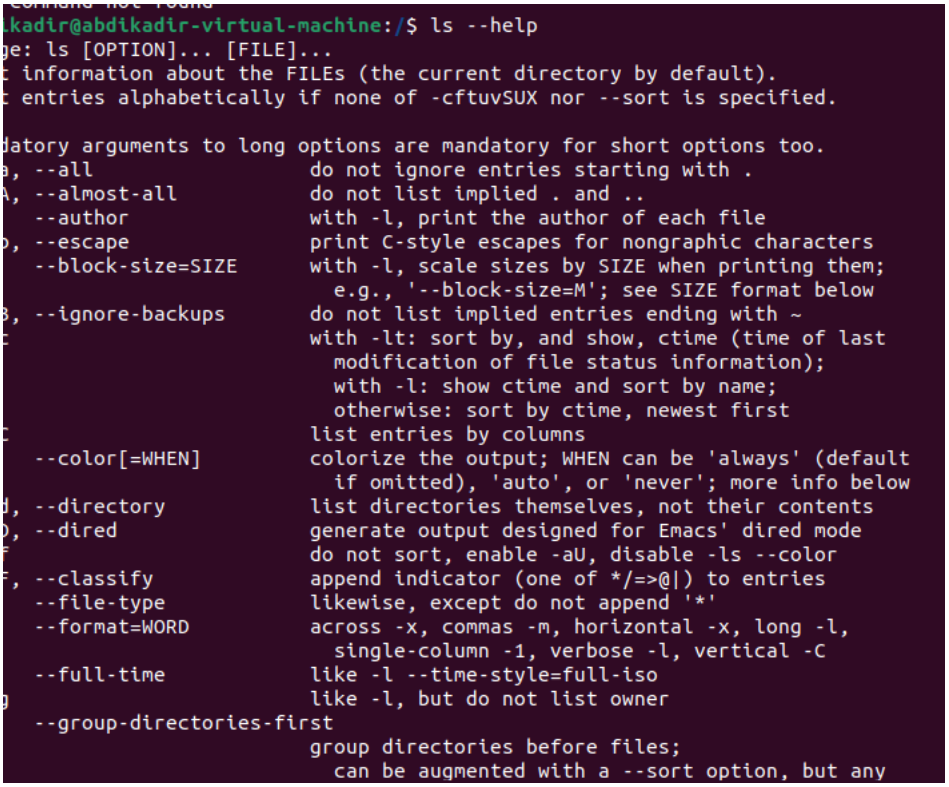
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ man rmdir
```

52. `command --help`

Definition: Most commands print a short help summary with --help. "command" stands for any command name.

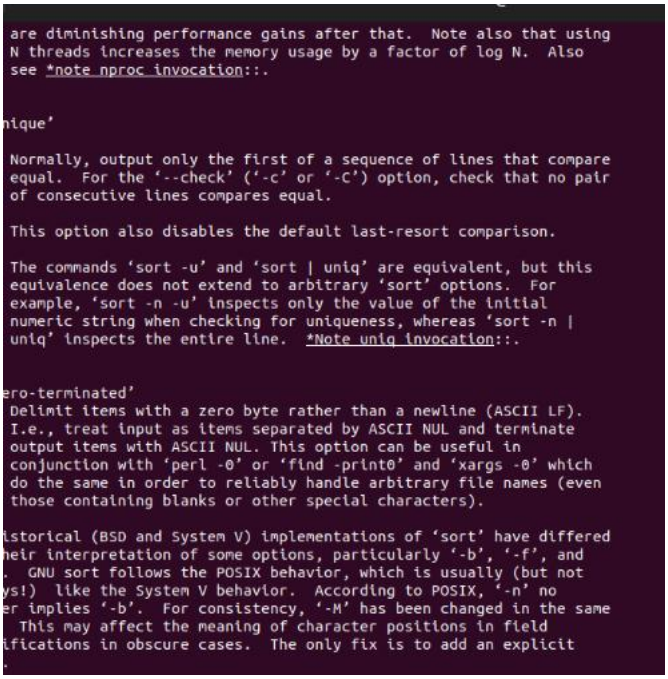
Test in terminal:



53. `info command`

Definition: Opens the GNU info documentation of a command. "command" stands for any command name. Press q to quit.

Test in terminal:



Creating files and folders

54. `cp /etc/hostname .`

Definition: Copies the hostname file from /etc into the current folder (the dot means here).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cp /etc/hostname .
```

55. `touch notes.txt todo.md`

Definition: Creates empty files (or updates their time if they exist).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ touch notes.txt todo.md
```

56. `mkdir week3`

Definition: Make Directory. Creates a new folder named week3.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mkdir week3
```

57. `mkdir -p projects/linux`

Definition: Creates a folder and any missing parent folders (-p).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mkdir -p projects/linux
```

58. `mkdir -p ~/ch2lab/notes`

Definition: Creates ch2lab and notes inside it in one command.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mkdir -p ~/ch2lab/notes
```

59. `mkdir ~/ch2lab/backup`

Definition: Creates the backup folder inside ch2lab.

Before testing: Run mkdir -p ~/ch2lab/notes first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mkdir ~/ch2lab/backup
```

60. `touch notes/day1.txt notes/day2.txt "notes/week 3.txt"`

Definition: Creates three files inside notes. Quotes are needed because "week 3.txt" has a space.

Before testing: Run mkdir notes first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ touch notes/day1.txt notes/day2.txt
"notes/week 3.txt"
```

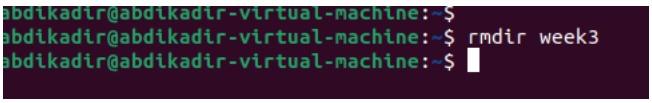
Removing folders (rmdir)

61. `rmdir week3`

Definition: Removes an EMPTY folder.

Before testing: Run mkdir week3 first.

Test in terminal:



62. `rmdir projects`

Definition: Fails if the folder is not empty. You will see "Directory not empty" because projects contains linux.

Before testing: Run mkdir -p projects/linux first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ rmdir projects
```

63. `rmdir notes-copy`

Definition: Removes the empty folder notes-copy.

Before testing: Run mkdir notes-copy first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ rmdir notes-copy
```

Copying (cp)

64. `cp notes.txt backup/`

Definition: Copies notes.txt into the backup folder.

Before testing: Run mkdir backup and touch notes.txt first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cp notes.txt backup/
```

65. `cp -r projects projects-copy`

Definition: Copies a whole folder with everything inside (-r means recursive).

Before testing: Run mkdir -p projects/linux first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cp -r projects projects-copy
```

66. `cp hostname hostname.bak`

Definition: Copies a file under a new name to make a backup.

Before testing: Run cp /etc/hostname . first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ touch notes.txt
abdikadir@abdikadir-virtual-machine:~$ cp notes.txt week3
abdikadir@abdikadir-virtual-machine:~$
```

67. `cp notes/day1.txt backup/`

Definition: Copies day1.txt from notes into backup.

Before testing: Create notes/day1.txt and backup first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cp notes/day1.txt backup/
```

68. `cp -r notes notes-copy`

Definition: Copies the whole notes folder to notes-copy.

Before testing: Create the notes folder first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cp -r notes notes-copy
```

Moving and renaming (mv)

69. `mv notes.txt old.txt`

Definition: Renames notes.txt to old.txt.

Before testing: Run touch notes.txt first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mv notes.txt old.txt
```

70. `mv draft.txt final.txt`

Definition: Renames draft.txt to final.txt.

Before testing: Run touch draft.txt first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mv draft.txt final.txt
```

71. `mv final.txt ~/Documents/`

Definition: Moves final.txt into the Documents folder.

Before testing: Run the previous mv first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mv final.txt ~/Documents/
```

72. `mv notes/day2.txt notes/monday.txt`

Definition: Renames a file inside the notes folder.

Before testing: Create notes/day2.txt first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ mv notes/day2.txt notes/monday.txt
```

Deleting (rm)

73. `rm FILE`

Definition: Deletes a file permanently (there is no recycle bin). Replace FILE with a real file name.

Before testing: Make sure old.txt exists.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ rm old.txt
```

74. `rm -i FILE`

Definition: Deletes a file but asks for confirmation first (-i = interactive).

Before testing: Make sure hostname.bak exists.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ rm -i hostname.bak
```

75. `rm -r DIR`

Definition: Deletes a folder and everything inside it (-r = recursive). Replace DIR with a real folder name.

Before testing: Make sure projects-copy exists.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ rm -r projects-copy
```

76. `rm -rf DIR`

Definition: Force-deletes a folder and its contents with no questions. Be very careful with this command. Use only on a test folder.

Before testing: Make sure notes-copy exists.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ rm -rf notes-copy
```

77. `rm -i notes.bak`

Definition: Asks before deleting the file notes.bak.

Before testing: Run touch notes.bak first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ rm -i notes.bak
```

78. `rm -i notes-copy/*`

Definition: Asks before deleting each file inside notes-copy (* means all files).

Before testing: Run cp -r notes notes-copy first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ rm -i notes-copy/*
```

Installing software

79. `sudo apt install sl`

Definition: Runs apt as administrator to install the small program sl. It needs internet in the VM and your password.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo apt install sl
```

80. `sl`

Definition: A fun program that shows a steam locomotive crossing the terminal. It is a joke for people who mistype ls. Install it first.

Before testing: Run sudo apt install sl first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sl
```

Chapter Three: Working with Text, Searching and Archives

Setup: create these practice files once before testing

```
abdikadir@abdikadir-virtual-machine:~$ cd ~/ch2lab
abdikadir@abdikadir-virtual-machine:~$ printf
"banana\napple\ncherry\napple\ndate\n" > fruits.txt
abdikadir@abdikadir-virtual-machine:~$ printf "red\ngreen\nblue\nyellow\nblack\n"
> colors.txt
abdikadir@abdikadir-virtual-machine:~$ printf "one:two:three\nfour:five:six\n" >
data.txt
```

Viewing file contents

81. `cat`

Definition: Shows the whole content of a file on the screen. It can also join several files together.

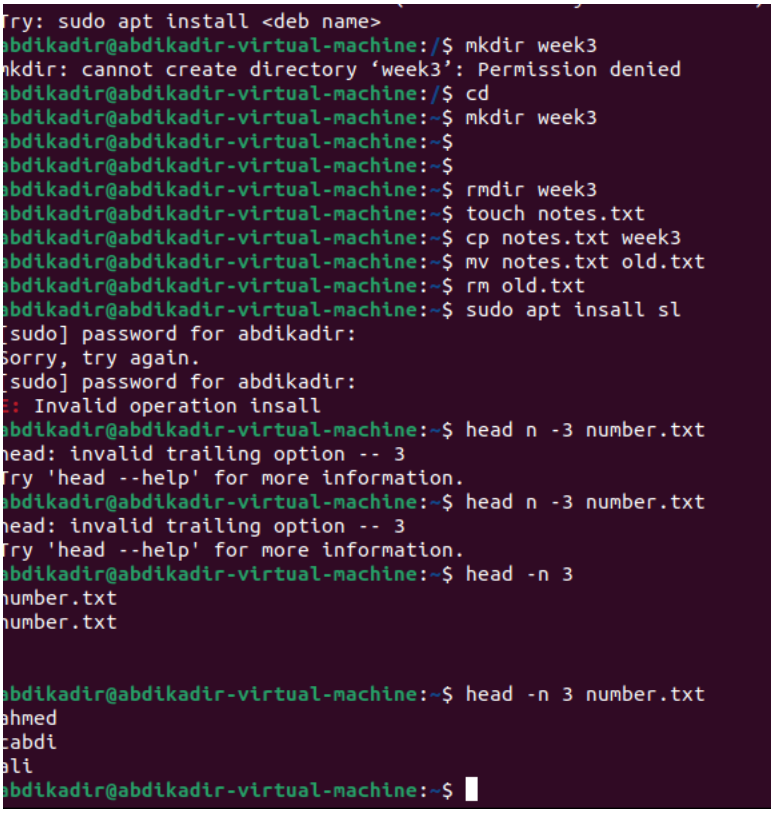
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cat fruits.txt
```

82. head

Definition: Shows the first lines of a file (10 lines by default). Use -n to choose how many.

Test in terminal:



83. `tail`

Definition: Shows the last lines of a file (10 lines by default). Use -n to choose how many.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ tail -n 3 fruits.txt
```

84. `less`

Definition: Opens a long file one page at a time. Use the arrow keys or Space to move, and q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ less /etc/services
```

Processing text

85. `wc`

Definition: Word Count. Counts the lines, words and bytes in a file. Use -l for lines only.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ wc fruits.txt
```

86. `sort`

Definition: Sorts the lines of a file in alphabetical order.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sort fruits.txt
```

87. `uniq`

Definition: Removes repeated lines that are next to each other. The file must be sorted first, so it is used after sort.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sort fruits.txt | uniq
```

88. `cut`

Definition: Cuts out one column (field) from each line. -d sets the separator and -f chooses the field number.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cut -d: -f1 data.txt
```

89. `paste`

Definition: Joins files side by side, line by line.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ paste fruits.txt colors.txt
```

90. `tr`

Definition: Translate. Replaces or deletes characters. The example changes small letters to capital letters.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ echo hello abdikadir | tr a-z A-Z
```

Searching

91. `grep`

Definition: Searches for a word or pattern inside files and prints the lines that match.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ grep apple fruits.txt
```

92. `find`

Definition: Searches for files and folders by name, type or size, starting from a folder you choose.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ find ~/ch2lab -name "*.txt"
```

93. `locate`

Definition: Finds files very quickly by searching a ready-made database of file names. If it is missing, install it and update the database first.

Before testing: If "command not found": sudo apt install plocate, then sudo updatedb.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ locate hostname
```

94. `which`

Definition: Shows the full path of the program that runs when you type a command.

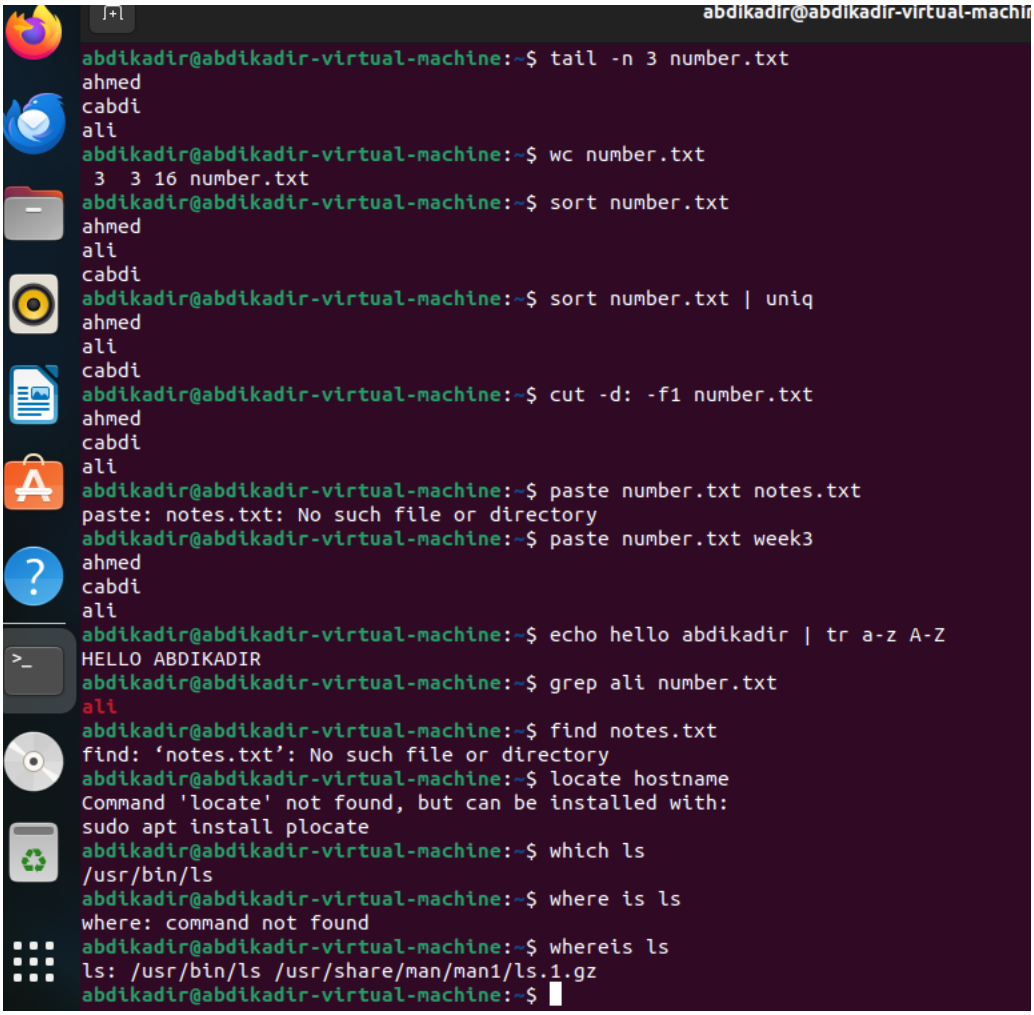
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ which ls
```

95. `whereis`

Definition: Shows where the program file, its source and its manual page are located.

Test in terminal:



Compressing and archiving

96. `tar`

Definition: Puts many files into one archive file. -c creates, -v shows progress, -f names the file. Use -t to list what is inside.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ tar -cvf backup.tar fruits.txt colors.txt
data.txt
```

97. `gzip`

Definition: Compresses a file to make it smaller and adds the .gz ending. -k keeps the original file.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ gzip -k colors.txt
```

98. `gunzip`

Definition: Uncompresses a .gz file back to the original.

Before testing: Run: cp colors.txt c2.txt && gzip c2.txt

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ gunzip c2.txt.gz
```

99. `zip`

Definition: Compresses files into a .zip archive that also works on Windows.

Before testing: If "command not found": sudo apt install zip unzip.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ zip files.zip fruits.txt data.txt
```

100. `unzip`

Definition: Extracts the files from a .zip archive. -d chooses the folder to extract into.

Before testing: Run the zip command first.

Test in terminal:

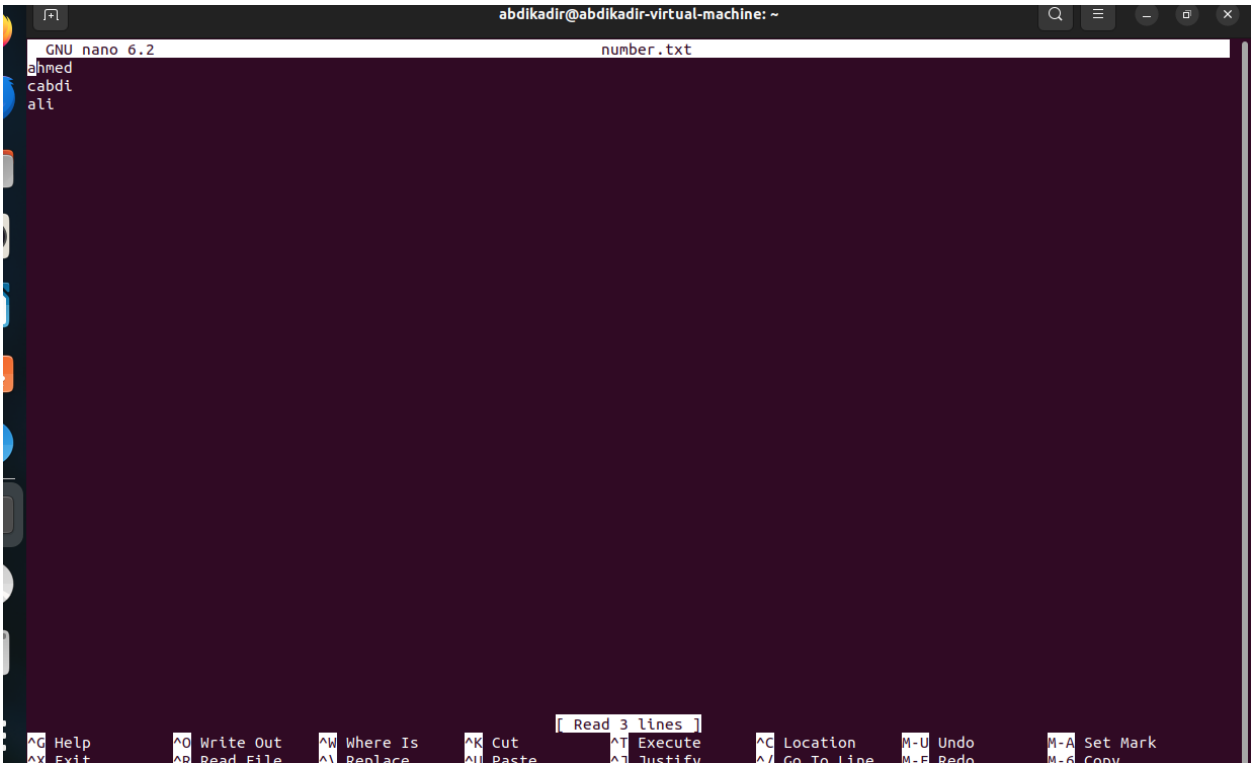
```
abdikadir@abdikadir-virtual-machine:~$ unzip files.zip -d unzipped
```

Text editors

101. nano

Definition: A simple terminal text editor. Type your text, press Ctrl + O then Enter to save, and Ctrl + X to exit.

Test in terminal:



102. `vim`

Definition: A powerful terminal editor. Press i to type, Esc to stop typing, then :wq and Enter to save and quit (or :q! to quit without saving).

Before testing: If "command not found": sudo apt install vim.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ vim notes.txt
```

Chapter Four: Processes, System Information, Services and Logs

Notes: PID, UNIT, N, USER, NAME and MESSAGE are placeholders. Each test command already uses a real value (for example cron as the service and abdikadir as the user). Run shutdown, reboot and poweroff last because they close the VM.

Viewing processes (ps)

103. `ps`

Definition: Process Status. Shows the processes running in your current terminal.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps
```

104. `ps -f`

Definition: Full format. Adds columns such as the user, parent PID (PPID) and start time.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -f
```

105. `ps -p PID`

Definition: Shows one process by its PID (process ID number). PID 1 is the first process of the system.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -p 1
```

106. `ps -o pid,ppid,comm`

Definition: Chooses your own columns: process ID, parent ID and command name.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -o pid,ppid,comm
```

107. `ps -p 1 -o pid,comm`

Definition: Shows only the PID and command name of process 1.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -p 1 -o pid,comm
```

108. `ps -p 1 -o pid,user,comm`

Definition: Shows the PID, the owner (user) and the command name of process 1.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -p 1 -o pid,user,comm
```

109. `ps aux`

Definition: Shows every process of all users, with CPU and memory use.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps aux
```

110. `ps aux | less`

Definition: Same as ps aux but one page at a time. Press q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps aux | less
```

111. `ps aux --sort=-%mem | head -n 5`

Definition: Shows the 5 processes that use the most memory (the - means biggest first).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps aux --sort=-%mem | head -n 5
```

112. `ps -ef`

Definition: Shows every process in full format (another common style of ps aux).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -ef
```

113. `ps -ef | head -n 4`

Definition: Shows only the first 4 lines of the full process list.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -ef | head -n 4
```

114. `ps -ef | less`

Definition: Shows the full process list one page at a time. Press q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -ef | less
```

115. `ps -o pid,ni,comm -p PID`

Definition: Shows the PID, the nice value (priority) and the name of one process. $$ is the PID of your own shell.

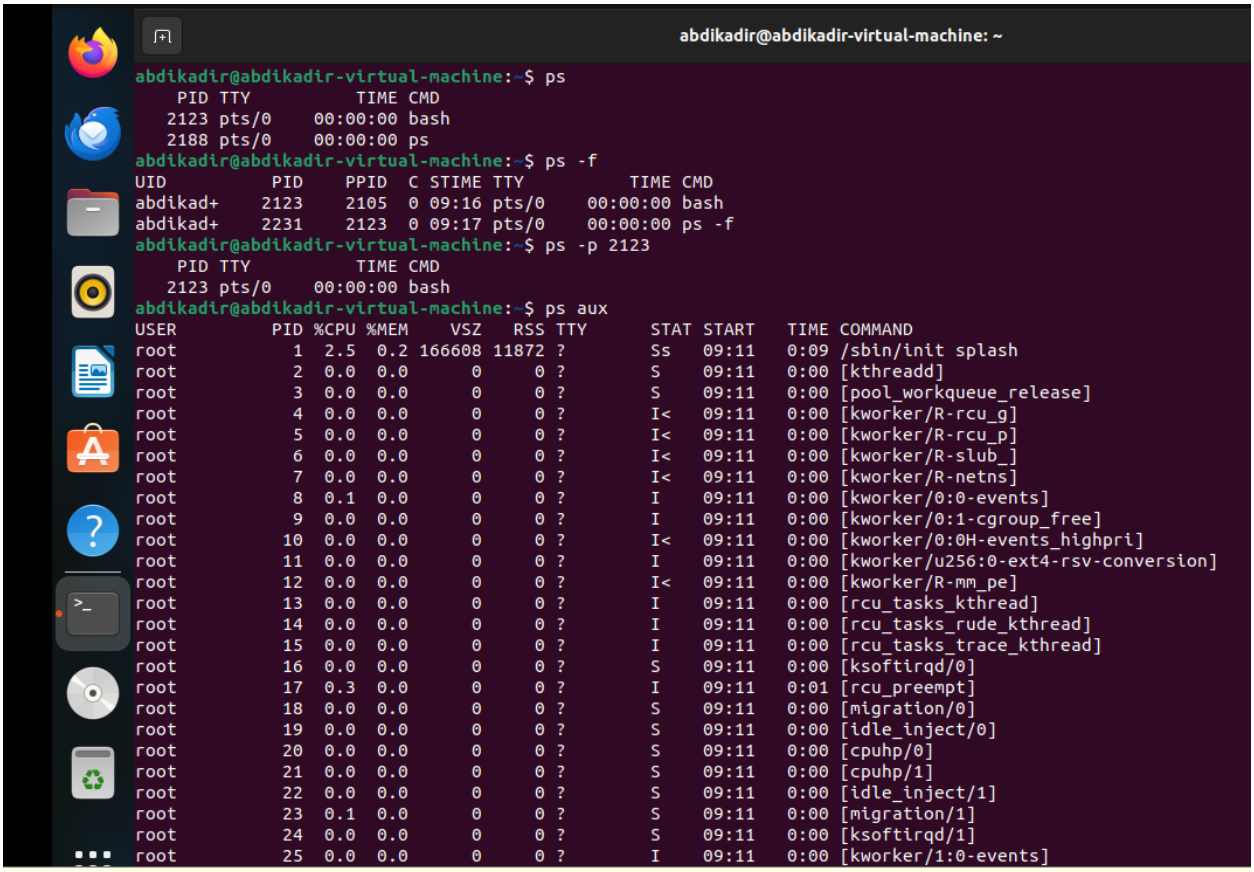
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ps -o pid,ni,comm -p $$
```

116. `ps -p PID -o pid,ppid,stat,cmd`

Definition: Shows PID, parent PID, state (stat) and the full command of one process.

Test in terminal:



Process trees and live monitoring

117. `pstree -p`

Definition: Shows all processes as a tree (parents and children) with their PIDs.

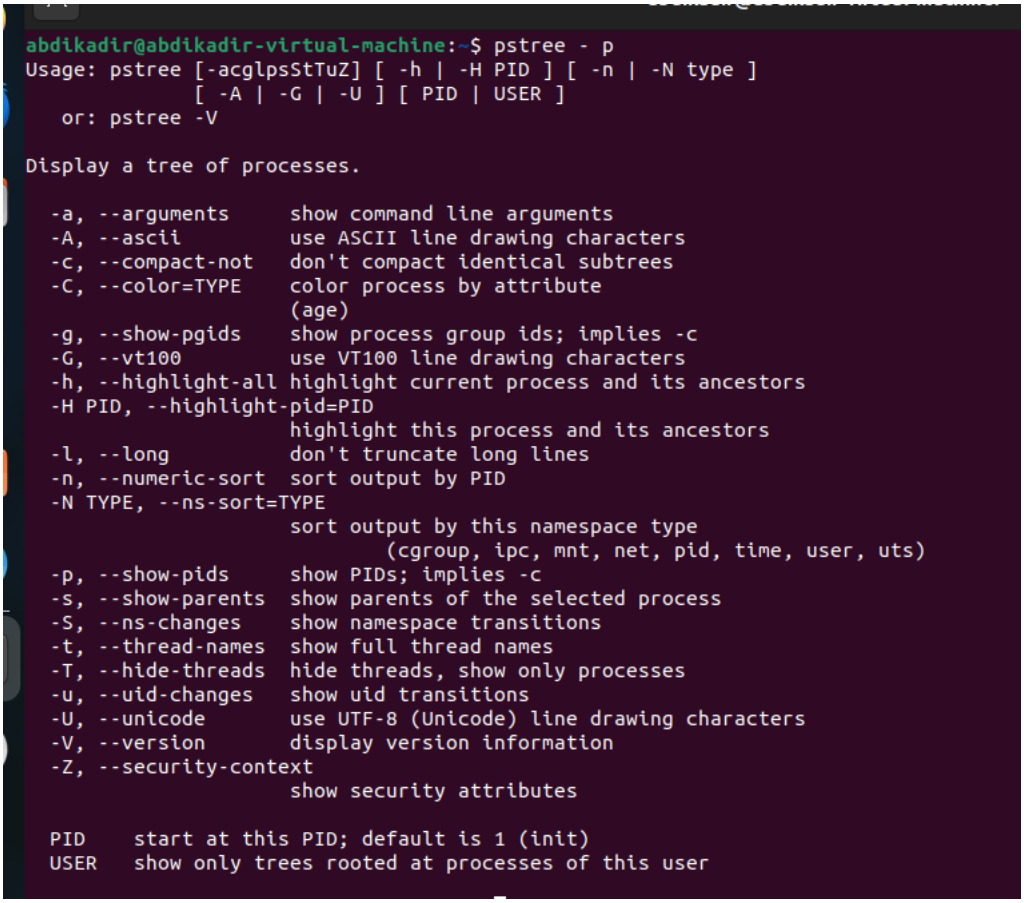
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ pstree -p
```

118. `pstree -p PID`

Definition: Shows the process tree that starts at one process.

Test in terminal:



119. `top`

Definition: Shows a live, updating list of processes with CPU and memory use. Press q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ top
```

120. `top -u USER`

Definition: Shows live processes of one user only. Use your own username.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ top -u abdikadir
```

121. `top -p PID`

Definition: Shows live information for one process only. Press q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ top -p 1
```

122. `htop`

Definition: A colourful and easier version of top. Use F10 or q to quit.

Before testing: If "command not found": sudo apt install htop.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ htop
```

123. `htop -u USER`

Definition: Shows htop for one user only.

Before testing: Install htop first if needed.

Test in terminal:

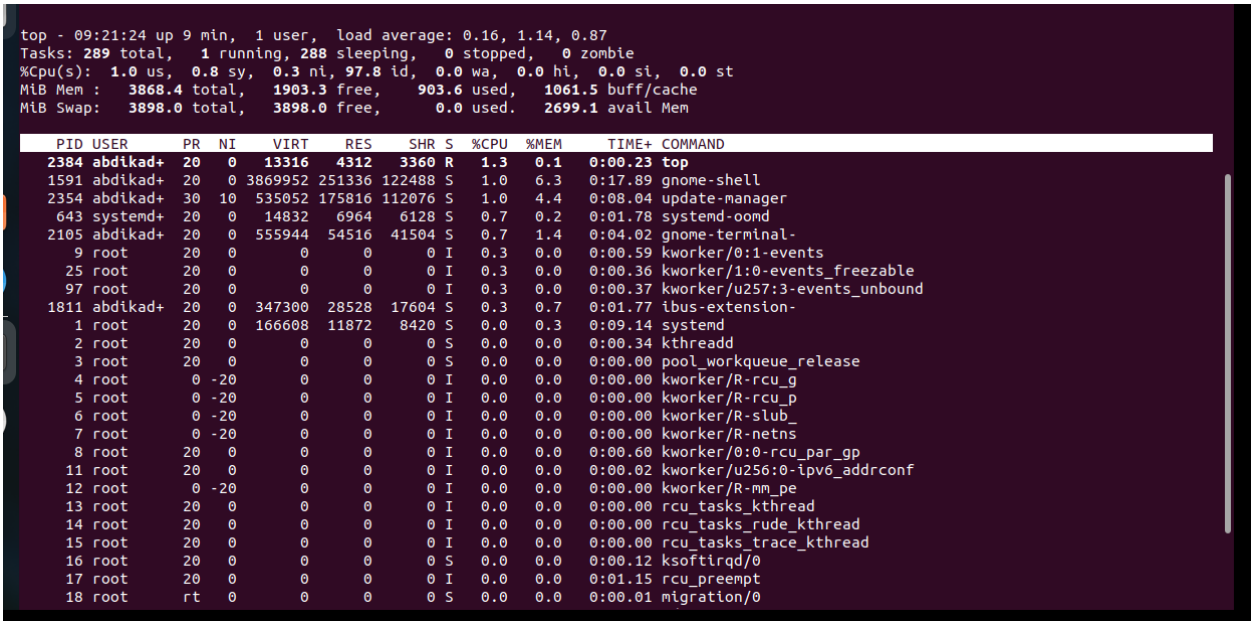
```
abdikadir@abdikadir-virtual-machine:~$ htop -u abdikadir
```

124. `htop -t`

Definition: Starts htop in tree view, so you can see which process started which.

Before testing: Install htop first if needed.

Test in terminal:



125. `command -v htop`

Definition: Checks if htop is installed. It prints the path if found and prints nothing if not found.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ command -v htop
```

Foreground and background jobs

126. `sleep 30`

Definition: Waits for 30 seconds and does nothing. It runs in the foreground, so the terminal is busy until it finishes (Ctrl + C stops it).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sleep 30
```

127. `sleep 30 &`

Definition: The & symbol runs the command in the background, so you get the prompt back at once.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sleep 30 &
```

128. `sleep 100`

Definition: Runs a 100 second wait in the foreground. Press Ctrl + Z to pause it and send it away from the foreground.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sleep 100
```

129. `sleep 600 &`

Definition: Starts a 10 minute wait in the background. Useful for testing the jobs and kill commands.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sleep 600 &
```

130. `gnome-calculator`

Definition: Opens the calculator app. In the foreground, the terminal is busy until you close the calculator.

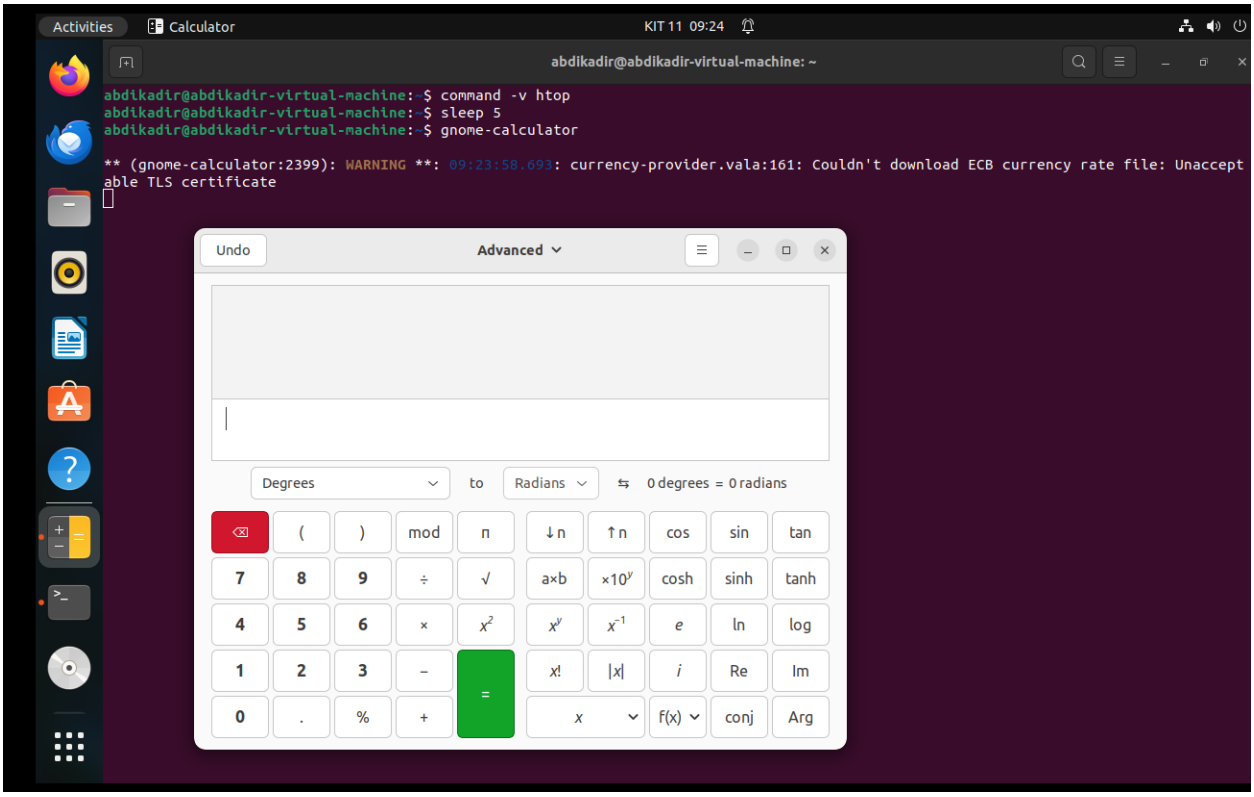
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ gnome-calculator
```

131. `gnome-calculator &`

Definition: Opens the calculator in the background, so the terminal stays free.

Test in terminal:



132. `jobs`

Definition: Lists the background and paused jobs of the current terminal.

Before testing: Run sleep 600 & first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ jobs
```

133. `jobs -l`

Definition: Lists jobs together with their PIDs.

Before testing: Run sleep 600 & first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ jobs -l
```

134. `bg`

Definition: Background. Continues the last paused job in the background.

Before testing: Run sleep 100, then press Ctrl + Z first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ bg
```

135. `bg %N`

Definition: Continues a paused job in the background by its job number. N is the number shown by jobs.

Before testing: Run sleep 100, press Ctrl + Z, then run jobs.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ bg %1
```

136. `fg`

Definition: Foreground. Brings the last background job back to the foreground.

Before testing: Run sleep 600 & first. Press Ctrl + C to stop it afterwards.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ fg
```

137. `fg %N`

Definition: Brings a job back to the foreground by its job number.

Before testing: Run sleep 600 & first. Press Ctrl + C to stop it afterwards.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ fg %1
```

Signals and stopping processes

138. `kill -l`

Definition: Lists all the signal names and numbers that kill can send.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ kill -l
```

139. `kill -l 15`

Definition: Shows the name of signal number 15 (it is TERM).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ kill -l 15
```

140. `kill PID`

Definition: Sends the TERM signal, which asks a process to stop politely. $! is the PID of the last background job.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sleep 600 & kill $!
```

141. `kill -TERM PID`

Definition: Same as kill PID, but writes the signal name (TERM) clearly.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sleep 600 & kill -TERM $!
```

142. `kill -HUP PID`

Definition: Sends the HUP (hang up) signal. Many programs reload their settings when they get it.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sleep 600 & kill -HUP $!
```

143. `kill %N`

Definition: Stops a background job by its job number.

Before testing: Run sleep 600 & first, then jobs to see the number.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ kill %1
```

144. `kill -9 PID`

Definition: Sends the KILL signal, which forces a process to stop right away. Use it only when kill PID fails.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sleep 600 & kill -9 $!
```

145. `killall NAME`

Definition: Stops all processes that have the given name.

Before testing: Run sleep 600 & first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ killall sleep
```

146. `pkill NAME`

Definition: Stops processes by matching part of their name.

Before testing: Run sleep 600 & first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ pkill sleep
```

147. `pkill -u USER NAME`

Definition: Stops processes with that name that belong to one user.

Before testing: Run sleep 600 & first.

Test in terminal:

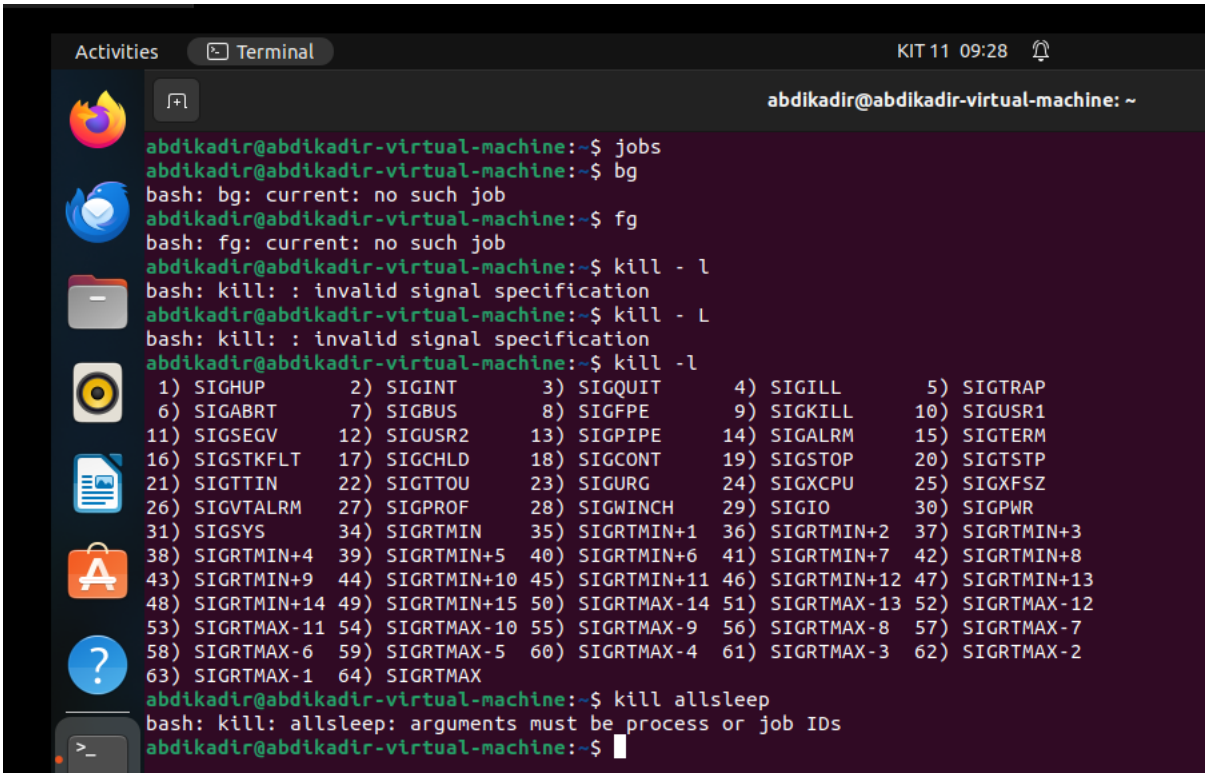
```
abdikadir@abdikadir-virtual-machine:~$ pkill -u abdikadir sleep
```

148. `pgrep -a NAME`

Definition: Finds processes by name and shows their PID with the full command (-a).

Before testing: Run sleep 600 & first.

Test in terminal:



Process priority (nice)

149. `nice`

Definition: With no arguments, it prints the current nice value (priority). The default is 0.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ nice
```

150. `nice -n N command`

Definition: Starts a command with a chosen nice value N. A higher number means lower priority.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ nice -n 5 ls
```

151. `nice -n 10 sleep 300 &`

Definition: Starts sleep in the background with a low priority (nice value 10).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ nice -n 10 sleep 300 &
```

152. `renice -n N -p PID`

Definition: Changes the nice value of a running process. Normal users can only increase the number.

Before testing: Run nice -n 10 sleep 300 & first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ renice -n 15 -p $(pgrep -n sleep)
```

153. `sudo renice -n N -p PID`

Definition: Same as renice, but as administrator. This lets you use a negative number to give more priority.

Before testing: Run nice -n 10 sleep 300 & first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo renice -n -5 -p $(pgrep -n sleep)
```

System information

154. `uname`

Definition: Prints the name of the operating system kernel (Linux).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ uname
```

155. `uname -r`

Definition: Prints the kernel release (version).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ uname -r
```

156. `uname -m`

Definition: Prints the machine hardware type, for example x86_64.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ uname -m
```

157. `uname -n`

Definition: Prints the computer (host) name.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ uname -n
```

158. `uname -a`

Definition: Prints all the system information together.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ uname -a
```

159. `hostnamectl`

Definition: Shows the host name, operating system, kernel and virtualization of the system.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ hostnamectl
```

160. `hostnamectl hostname`

Definition: Prints only the host name.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ hostnamectl hostname
```

161. `uptime`

Definition: Shows the current time, how long the system has been running, the number of users and the load average.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ uptime
```

162. `uptime -p`

Definition: Shows the uptime in a pretty, easy to read form.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ uptime -p
```

163. `uptime -s`

Definition: Shows the date and time the system started.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ uptime -s
```

164. `nproc`

Definition: Prints the number of CPU processing units available.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ nproc
```

165. `free -h`

Definition: Shows memory (RAM) and swap use in human-readable sizes.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ free -h
```

166. `free -m`

Definition: Shows memory use in megabytes.

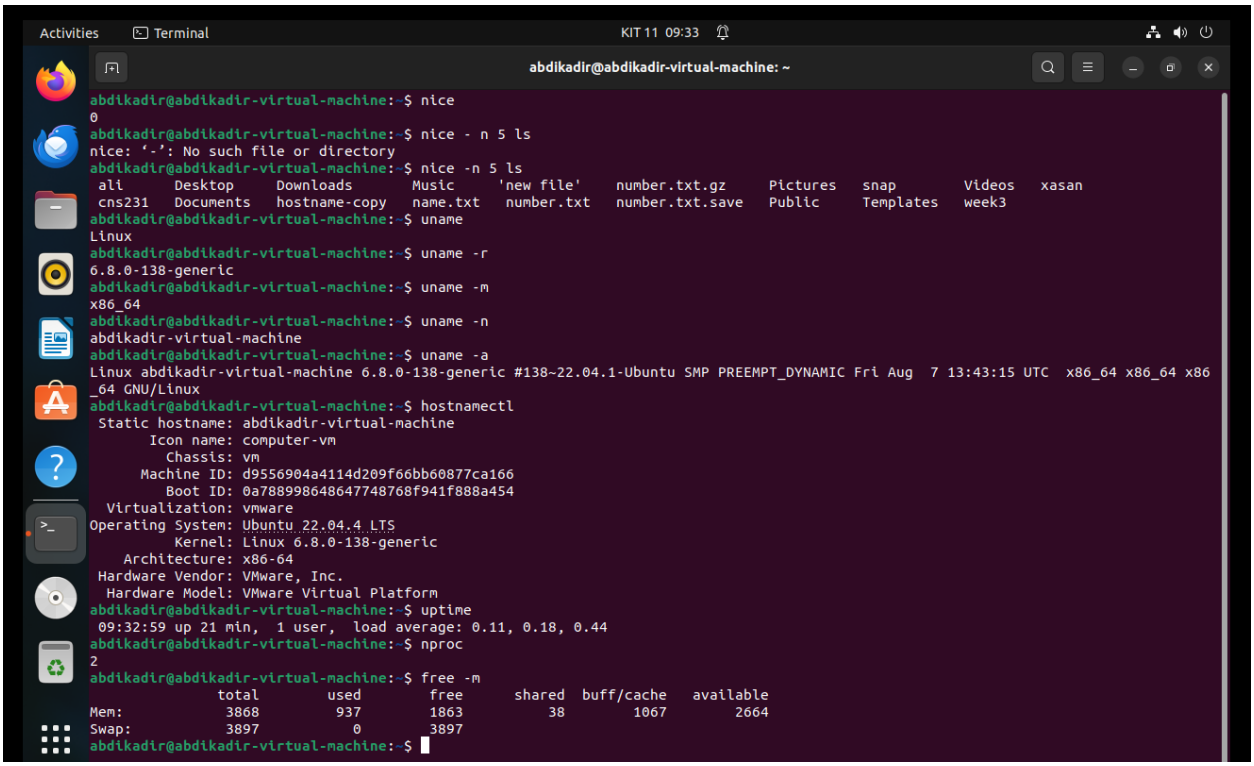
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ free -m
```

167. `free -h -s N`

Definition: Repeats the memory report every N seconds. Press Ctrl + C to stop.

Test in terminal:



168. `lscpu`

Definition: Shows information about the CPU, such as model, cores and speed.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ lscpu
```

169. `lscpu | less`

Definition: Shows the CPU information one page at a time. Press q to quit.

Test in terminal:



170. `lsblk`

Definition: Lists block devices, which are the disks and partitions.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ lsblk
```

171. `lsblk -e 7`

Definition: Lists disks and partitions but hides loop devices (type 7).

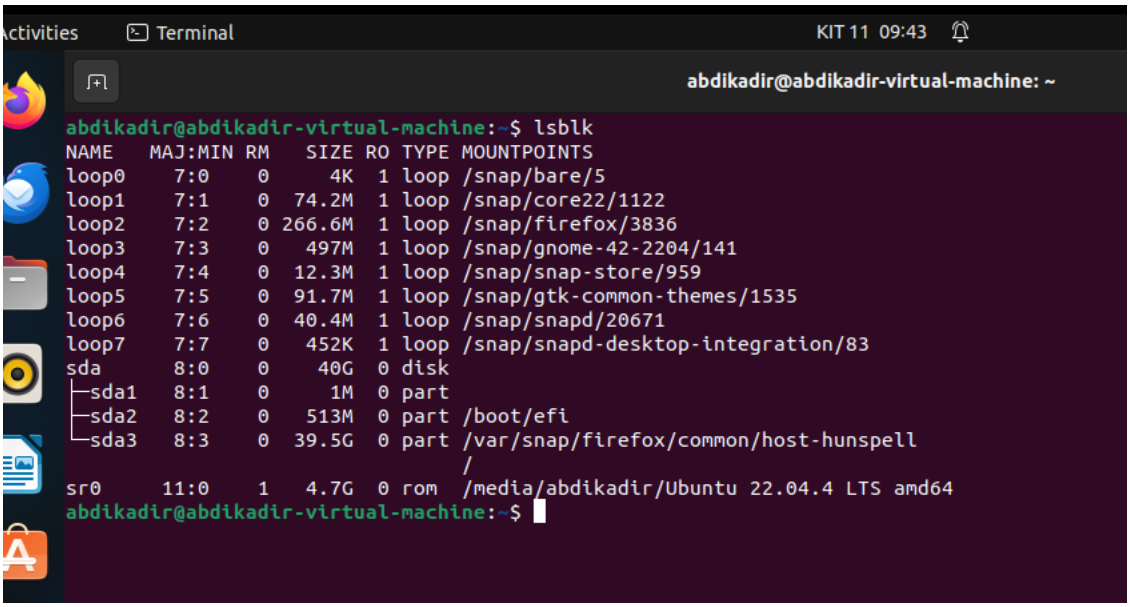
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ lsblk -e 7
```

172. `lsblk -e 7 -d -o NAME,SIZE,MODEL`

Definition: Shows only whole disks (-d) with the columns name, size and model.

Test in terminal:



The /proc folder

173. `ls /proc`

Definition: Lists /proc, a special folder where Linux shows live information about the system and processes. Numbers are process IDs.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls /proc
```

174. `ls -l /proc/meminfo`

Definition: Shows the memory information file in long format. Its size is 0 because the system makes it live.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls -l /proc/meminfo
```

175. `cat /proc/uptime`

Definition: Shows how many seconds the system has been running.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cat /proc/uptime
```

176. `cat /proc/loadavg`

Definition: Shows the system load average for 1, 5 and 15 minutes.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ cat /proc/loadavg
```

177. `ls /proc/$$`

Definition: Lists the information files of your own shell process ($$ is its PID).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ ls /proc/$$
```

178. `head -n 7 /proc/$$/status`

Definition: Shows the first 7 lines of the status of your shell process.

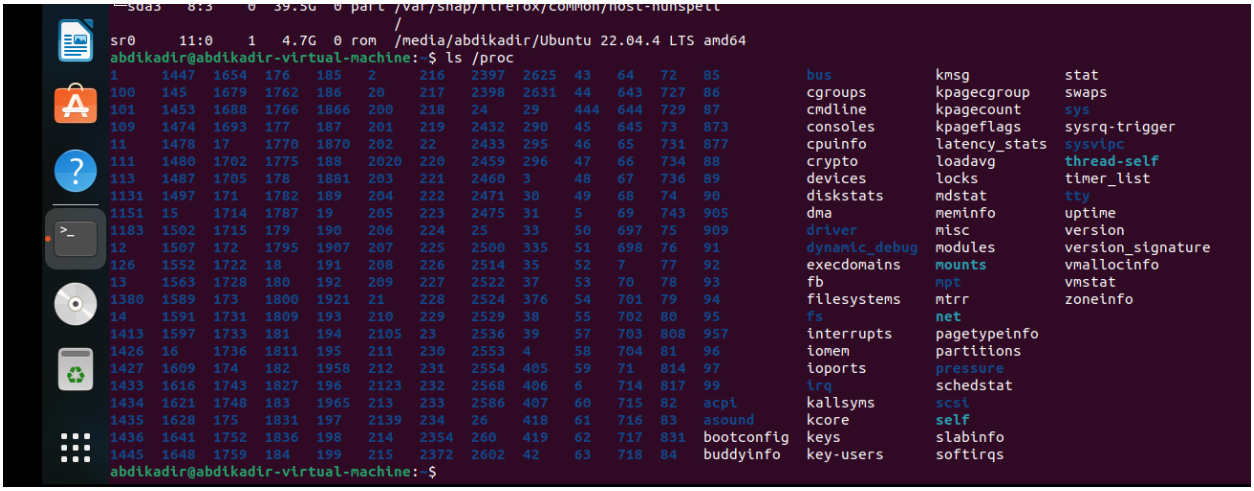
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ head -n 7 /proc/$$/status
```

179. `cat /proc/1/comm`

Definition: Shows the name of process 1 (usually systemd).

Test in terminal:



180. `grep -m 1 "model name" /proc/cpuinfo`

Definition: Shows the CPU model name. -m 1 stops after the first match.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ grep -m 1 "model name" /proc/cpuinfo
```

181. `grep -c ^processor /proc/cpuinfo`

Definition: Counts how many processors the system has (-c counts the matching lines).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ grep -c ^processor /proc/cpuinfo
```

182. `head -n 3 /proc/meminfo`

Definition: Shows the first 3 lines of the memory information (total, free and available).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ head -n 3 /proc/meminfo
```

Services (systemctl)

183. `systemctl status UNIT`

Definition: Shows if a service is running, with its recent log lines. UNIT is the service name (for example cron). Press q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ systemctl status cron
```

184. `systemctl is-active UNIT`

Definition: Prints active or inactive for a service.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ systemctl is-active cron
```

185. `systemctl is-enabled UNIT`

Definition: Prints enabled or disabled, which tells if the service starts at boot.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ systemctl is-enabled cron
```

186. `sudo systemctl start UNIT`

Definition: Starts a service now.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl start cron
```

187. `sudo systemctl stop UNIT`

Definition: Stops a service now.

Before testing: Start it again afterwards: sudo systemctl start cron.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl stop cron
```

188. `sudo systemctl restart UNIT`

Definition: Stops and starts a service again.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl restart cron
```

189. `systemctl show -p MainPID UNIT`

Definition: Shows one property of a service, here the PID of its main process.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ systemctl show -p MainPID cron
```

190. `sudo systemctl reload UNIT`

Definition: Asks a service to reload its settings without stopping. Only services that support reload accept it.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl reload NetworkManager
```

191. `sudo systemctl reload-or-restart UNIT`

Definition: Reloads the service if it supports reload. If not, it restarts it.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl reload-or-restart cron
```

192. `sudo systemctl enable UNIT`

Definition: Makes a service start automatically at boot.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl enable cron
```

193. `sudo systemctl disable UNIT`

Definition: Stops a service from starting automatically at boot.

Before testing: Enable it again afterwards: sudo systemctl enable cron.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl disable cron
```

194. `sudo systemctl enable --now UNIT`

Definition: Enables a service at boot and starts it right now.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl enable --now cron
```

195. `sudo systemctl disable --now UNIT`

Definition: Disables a service at boot and stops it right now.

Before testing: Enable it again afterwards: sudo systemctl enable --now cron.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo systemctl disable --now cron
```

196. `systemctl list-units --type=service --state=running`

Definition: Lists all services that are running now.

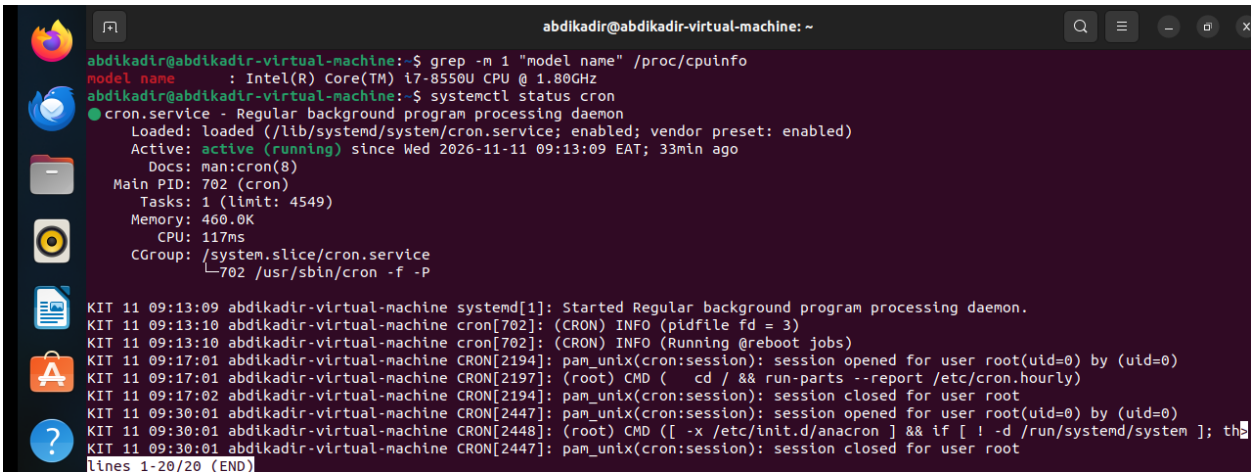
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ systemctl list-units --type=service --
state=running
```

197. `systemctl list-units --type=service --state=failed`

Definition: Lists services that failed to start. It may show 0 loaded units, which is a good result.

Test in terminal:



System logs (journalctl)

198. `journalctl`

Definition: Shows the system journal (all logs), oldest first. Press q to quit.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl
```

199. `journalctl -n N`

Definition: Shows only the last N log lines.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -n 10
```

200. `journalctl -b`

Definition: Shows logs from the current boot only.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -b
```

201. `journalctl -b -p err`

Definition: Shows only error messages (and worse) from the current boot.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -b -p err
```

202. `journalctl -u UNIT -n N`

Definition: Shows the last N log lines of one service.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -u cron -n 10
```

203. `journalctl -u UNIT --since "1 hour ago"`

Definition: Shows the logs of one service from the last hour.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -u cron --since "1 hour ago"
```

204. `journalctl -u UNIT -n 20`

Definition: Shows the last 20 log lines of one service.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -u cron -n 20
```

205. `journalctl -xe`

Definition: Jumps to the end of the journal with extra explanations (x). Useful after a service fails.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -xe
```

206. `journalctl -f`

Definition: Follows the journal live, showing new lines as they arrive. Press Ctrl + C to stop.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -f
```

207. `journalctl -u UNIT -f`

Definition: Follows the live logs of one service. Press Ctrl + C to stop.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl -u cron -f
```

208. `journalctl _COMM=sudo -n 1`

Definition: Shows the last log line made by the sudo command.

Before testing: Run any sudo command first so a line exists.

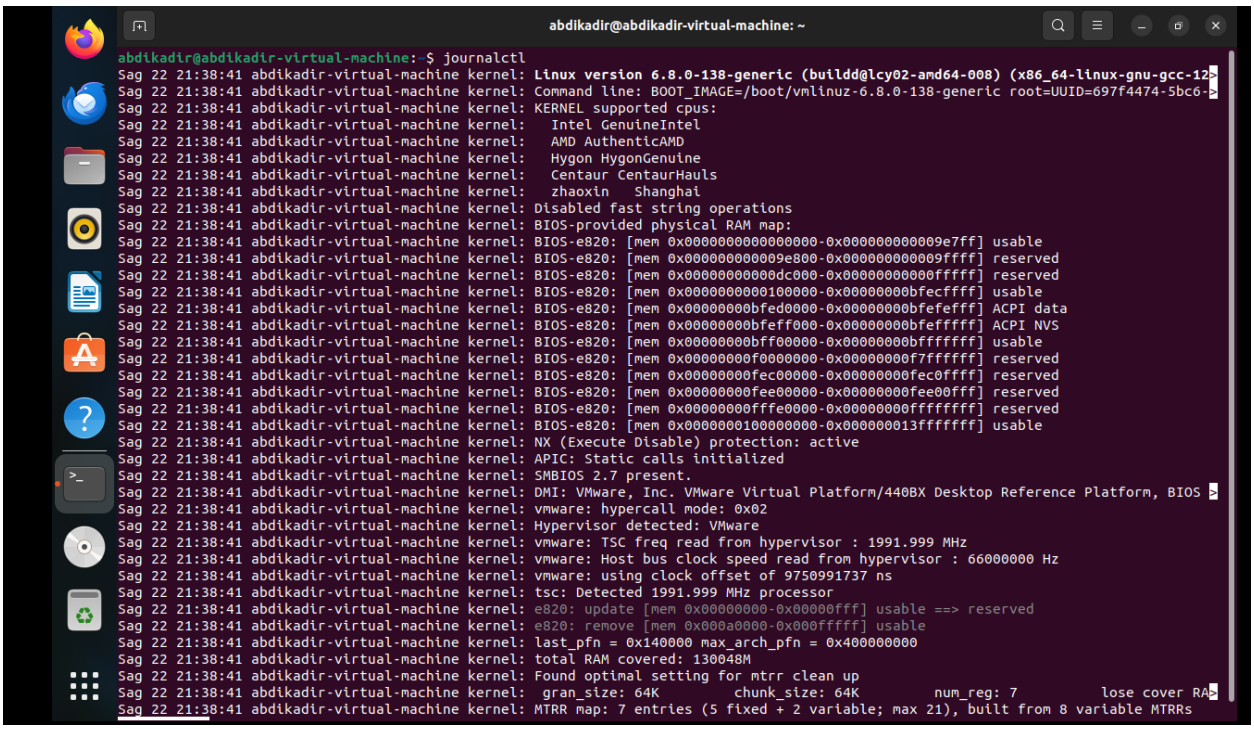
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ journalctl _COMM=sudo -n 1
```

209. `journalctl -k -b`

Definition: Shows kernel messages from the current boot.

Test in terminal:



Log files in /var/log

210. `less /var/log/syslog`

Definition: Opens the main system log file one page at a time. Press q to quit.

Before testing: If the file is missing, use journalctl instead.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ less /var/log/syslog
```

211. `tail -n 3 /var/log/kern.log`

Definition: Shows the last 3 lines of the kernel log file.

Before testing: If the file is missing, use journalctl -k instead.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ tail -n 3 /var/log/kern.log
```

212. `tail -f /var/log/syslog`

Definition: Follows the system log live. Press Ctrl + C to stop.

Before testing: If the file is missing, use journalctl -f instead.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ tail -f /var/log/syslog
```

213. `grep sudo /var/log/auth.log`

Definition: Finds all lines about sudo in the login and security log. Administrator rights are needed to read this file.

Before testing: If the file is missing, use journalctl _COMM=sudo instead.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo grep sudo /var/log/auth.log
```

214. `grep sudo /var/log/auth.log | tail -n 1`

Definition: Shows only the last sudo line from the security log.

Before testing: If the file is missing, use journalctl _COMM=sudo -n 1 instead.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo grep sudo /var/log/auth.log | tail -n
1
```

215. `last -n 2`

Definition: Shows the last 2 logins to the system.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ last -n 2
```

216. `sudo dmesg -T`

Definition: Shows kernel messages (hardware and drivers) with readable time stamps (-T).

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo dmesg -T
```

217. `sudo dmesg -T | tail -n 2`

Definition: Shows only the last 2 kernel messages.

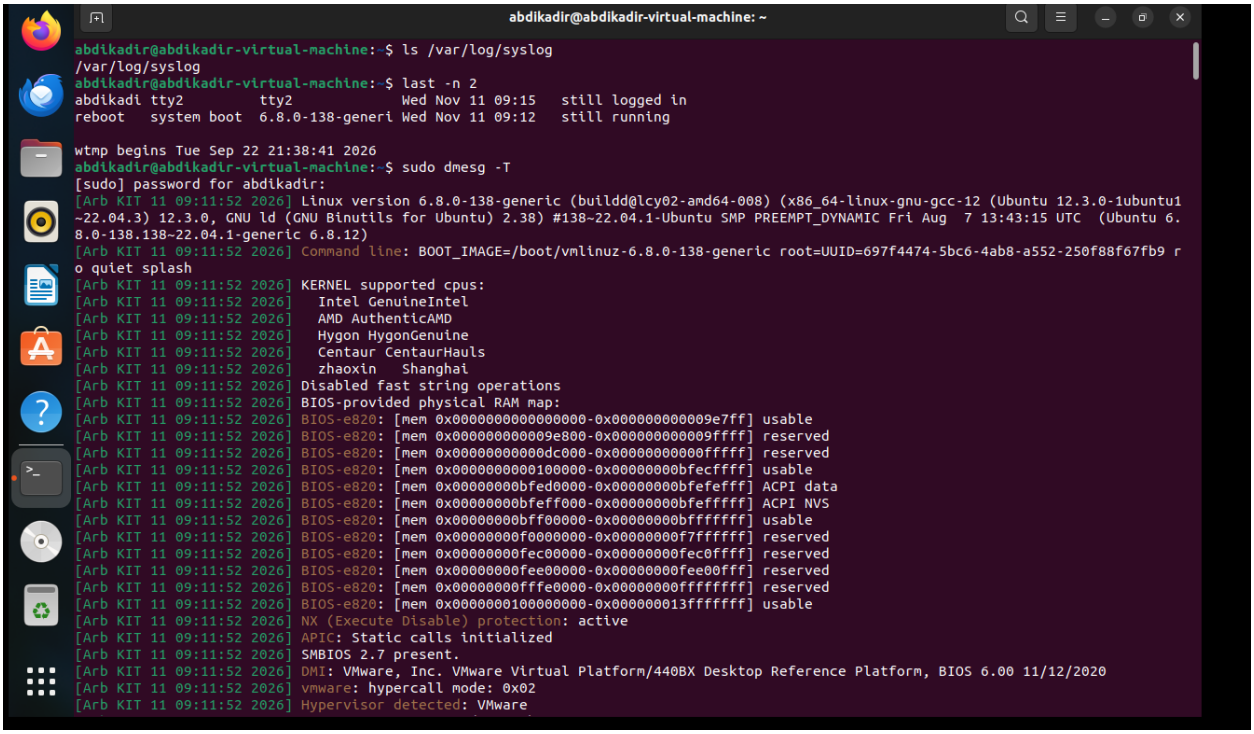
Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo dmesg -T | tail -n 2
```

218. `sudo dmesg | less`

Definition: Shows the kernel messages one page at a time. Press q to quit.

Test in terminal:



Shutdown, SSH test and logging

219. `sudo shutdown now`

Definition: Turns the computer off immediately. Warning: this closes your Ubuntu VM. Do this one last, after all other work is saved.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo shutdown now
```

220. `sudo shutdown -r +N "MESSAGE"`

Definition: Restarts (-r) the computer after N minutes and shows a message to users. Cancel it with the next command.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo shutdown -r +5 "Restarting for a test"
```

221. `sudo shutdown -c`

Definition: Cancels a shutdown or restart that is waiting.

Before testing: Run the shutdown -r +5 command first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo shutdown -c
```

222. `sudo reboot`

Definition: Restarts the computer right now. Warning: this restarts your VM. Save your work first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo reboot
```

223. `sudo poweroff`

Definition: Turns the computer off right now. Warning: this closes your VM. Save your work first.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo poweroff
```

224. `sudo sshd -t`

Definition: Tests the SSH server configuration file for mistakes. No output means the file is correct.

Before testing: If "command not found": sudo apt install openssh-server.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ sudo sshd -t
```

225. `echo $$`

Definition: Prints the PID (process ID) of your current shell.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ echo $$
```

226. `logger "MESSAGE"`

Definition: Writes your message into the system log. You can find it again with journalctl -n 1.

Test in terminal:

```
abdikadir@abdikadir-virtual-machine:~$ logger "Hello from abdikadir"
```